

Career Roadmap: Software Developer Role

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Phase 1: Computer Science Core & Data Structures

Master Array, String, Linked List, Stack, Queue, Binary Tree, Graph algorithms in Python/C++/Java.

Phase 2: Database & System Fundamentals

Study Relational Databases (PostgreSQL/MySQL), Normalization, Indexing, and Operating System / Networking basics.

Phase 3: Web Application Architecture

Build RESTful APIs with FastAPI or Express.js. Frontend integration using React and responsive CSS.

Phase 4: Software Engineering & Git Workflow

Practice Git, GitHub Actions, Docker basics, unit testing, and design patterns. Build a full-stack capstone project.